

Suppose we are given a pair of functions f and F analytic in regions Ω and Ω' , respectively, with $\Omega \subset \Omega'$. If the two functions agree on the smaller set Ω , we say that F is an **analytic continuation** of f into the region Ω' . The corollary then guarantees that there can be only one such analytic continuation, since F is uniquely determined by f .

5 Further applications

We gather in this section various consequences of the results proved so far.

5.1 Morera's theorem

A direct application of what was proved here is a converse of Cauchy's theorem.

Theorem 5.1 *Suppose f is a continuous function in the open disc D such that for any triangle T contained in D*

$$\int_T f(z) dz = 0,$$

then f is holomorphic.

Proof. By the proof of Theorem 2.1 the function f has a primitive F in D that satisfies $F' = f$. By the regularity theorem, we know that F is indefinitely (and hence twice) complex differentiable, and therefore f is holomorphic.

5.2 Sequences of holomorphic functions

Theorem 5.2 *If $\{f_n\}_{n=1}^\infty$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω , then f is holomorphic in Ω .*

Proof. Let D be any disc whose closure is contained in Ω and T any triangle in that disc. Then, since each f_n is holomorphic, Goursat's theorem implies

$$\int_T f_n(z) dz = 0 \quad \text{for all } n.$$

By assumption $f_n \rightarrow f$ uniformly in the closure of D , so f is continuous and

$$\int_T f_n(z) dz \rightarrow \int_T f(z) dz.$$

As a result, we find $\int_T f(z) dz = 0$, and by Morera's theorem, we conclude that f is holomorphic in D . Since this conclusion is true for every D whose closure is contained in Ω , we find that f is holomorphic in all of Ω .

This is a striking result that is obviously not true in the case of real variables: the uniform limit of continuously differentiable functions need not be differentiable. For example, we know that every continuous function on $[0, 1]$ can be approximated uniformly by polynomials, by Weierstrass's theorem (see Chapter 5, Book I), yet not every continuous function is differentiable.

We can go one step further and deduce convergence theorems for the sequence of derivatives. Recall that if f is a power series with radius of convergence R , then f' can be obtained by differentiating term by term the series for f , and moreover f' has radius of convergence R . (See Theorem 2.6 in Chapter 1.) This implies in particular that if S_n are the partial sums of f , then S'_n converges uniformly to f' on every compact subset of the disc of convergence of f . The next theorem generalizes this fact.

Theorem 5.3 *Under the hypotheses of the previous theorem, the sequence of derivatives $\{f'_n\}_{n=1}^\infty$ converges uniformly to f' on every compact subset of Ω .*

Proof. We may assume without loss of generality that the sequence of functions in the theorem converges uniformly on all of Ω . Given $\delta > 0$, let Ω_δ denote the subset of Ω defined by

$$\Omega_\delta = \{z \in \Omega : \overline{D_\delta(z)} \subset \Omega\}.$$

In other words, Ω_δ consists of all points in Ω which are at distance $> \delta$ from its boundary. To prove the theorem, it suffices to show that $\{f'_n\}$ converges uniformly to f' on Ω_δ for each δ . This is achieved by proving the following inequality:

$$(13) \quad \sup_{z \in \Omega_\delta} |F'(z)| \leq \frac{1}{\delta} \sup_{\zeta \in \Omega} |F(\zeta)|$$

whenever F is holomorphic in Ω , since it can then be applied to $F = f_n - f$ to prove the desired fact. The inequality (13) follows at once from the Cauchy integral formula and the definition of Ω_δ , since for every $z \in \Omega_\delta$ the closure of $D_\delta(z)$ is contained in Ω and

$$F'(z) = \frac{1}{2\pi i} \int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta - z)^2} d\zeta.$$

Hence,

$$\begin{aligned}
 |F'(z)| &\leq \frac{1}{2\pi} \int_{C_\delta(z)} \frac{|F(\zeta)|}{|\zeta - z|^2} |d\zeta| \\
 &\leq \frac{1}{2\pi} \sup_{\zeta \in \Omega} |F(\zeta)| \frac{1}{\delta^2} 2\pi\delta \\
 &= \frac{1}{\delta} \sup_{\zeta \in \Omega} |F(\zeta)|,
 \end{aligned}$$

as was to be shown.

Of course, there is nothing special about the first derivative, and in fact under the hypotheses of the last theorem, we may conclude (arguing as above) that for every $k \geq 0$ the sequence of k^{th} derivatives $\{f_n^{(k)}\}_{n=1}^\infty$ converges uniformly to $f^{(k)}$ on every compact subset of Ω .

In practice, one often uses Theorem 5.2 to construct holomorphic functions (say, with a prescribed property) as a series

$$(14) \quad F(z) = \sum_{n=1}^{\infty} f_n(z).$$

Indeed, if each f_n is holomorphic in a given region Ω of the complex plane, and the series converges uniformly in compact subsets of Ω , then Theorem 5.2 guarantees that F is also holomorphic in Ω . For instance, various special functions are often expressed in terms of a converging series like (14). A specific example is the Riemann zeta function discussed in Chapter 6.

We now turn to a variant of this idea, which consists of functions defined in terms of integrals.

5.3 Holomorphic functions defined in terms of integrals

As we shall see later in this book, a number of other special functions are defined in terms of integrals of the type

$$f(z) = \int_a^b F(z, s) ds,$$

or as limits of such integrals. Here, the function F is holomorphic in the first argument, and continuous in the second. The integral is taken in the sense of Riemann integration over the bounded interval $[a, b]$. The problem then is to establish that f is holomorphic.